

Lab 2D: Package and Ship - Plugins & Marketplaces

Duration: 35 min

After: Plugins & Marketplaces lecture (Day 2, Block 4)

Prerequisites: Labs 2A-2C complete (skills and hooks in `.claude/`)

Objective

Inspect a marketplace entry before installation, scaffold a skills-directory plugin with `claude plugin init`, validate it strictly, publish the proven package through a local marketplace, and install it back at an intentional scope.

Deliverable (toolkit chain 8 of 12)

A validated `dashboard-toolkit` plugin containing two skills and two hooks, plus a local marketplace that can distribute it.

START MARKER: `/clear done; /skills shows code-review and review; both hooks passed Lab 2B.`

Exercise 1: Inspect Before Installing (6 min)

Open the plugin manager:

```
/plugin
```

Choose a plugin from `claude-plugins-official`, open its detail view, and inspect the publisher plus every contributed component. The official marketplace is a catalog; entries can have different publishers and maintenance models.

Use this review checklist:

1. Publisher, source repository, release activity
2. Skills, commands, agents, hooks, scripts, MCP/LSP servers, dependencies
3. Files, commands, credentials, and user privileges it can access
4. Network destinations, model/provider changes, and stored state
5. Version/update policy plus disable, uninstall, and rollback path

Record an **APPROVE**, **REJECT**, or **APPROVE WITH CONDITIONS** decision with one sentence of evidence.

Plugins are executable dependencies. Marketplace placement is not a security boundary.

Exercise 2: Scaffold the Plugin (6 min)

Exit Claude and run:

```
claude plugin init dashboard-toolkit \
--description "Review skills and deterministic enforcement hooks" \
--with skills hooks
```

Current behavior: the command creates `~/.claude/skills/dashboard-toolkit/` and loads it as `dashboard-toolkit@skills-dir` on the next session. It does not scaffold in the current directory.

Inspect and validate:

```

claude plugin validate ~/.claude/skills/dashboard-toolkit --strict
claude plugin list

```

Expected output marker: strict validation succeeds and the plugin appears from the `skills-dir` source.

Exercise 3: Package the Toolkit (8 min)

Start Claude in `business-dashboard` and run:

```

Update ~/.claude/skills/dashboard-toolkit so it contains:
1. My .claude/skills/code-review/ skill as skills/code-review/
2. My .claude/skills/review/ human-invoked skill as skills/review/
3. My post-edit-format.sh and pre-write-check.sh scripts under hooks/
4. hooks/hooks.json declaring the PostToolUse Write|Edit formatter and PreToolUse Write
   blocker using ${CLAUDE_PLUGIN_ROOT}-relative paths
5. A manifest description that names the skills and hooks
Remove placeholder example components that the scaffold created.
Do not add an MCP server, external network dependency, credential, or provider override.

```

Validate again:

```

claude plugin validate ~/.claude/skills/dashboard-toolkit --strict
claude plugin details dashboard-toolkit@skills-dir

```

Expected output marker: validation passes and the component inventory matches the intended package. Review the inventory before continuing.

Exercise 4: Publish Through a Local Marketplace (10 min)

Create this structure next to `business-dashboard`:

```

team-marketplace/
.claude-plugin/
marketplace.json
plugins/
dashboard-toolkit/
.claude-plugin/plugin.json
skills/
hooks/

```

Copy the validated plugin into `team-marketplace/plugins/dashboard-toolkit/`. Create `.claude-plugin/marketplace.json`:

```

{
  "name": "team-marketplace",
  "owner": { "name": "Training Team" },
  "plugins": [
    {
      "name": "dashboard-toolkit",
      "source": "../plugins/dashboard-toolkit",
      "description": "Review skills and deterministic enforcement hooks"
    }
  ]
}

```

From `business-dashboard`, run:

```
/plugin marketplace add ../team-marketplace
/plugin install dashboard-toolkit@team-marketplace
```

Choose **local scope** for the classroom smoke test unless the instructor explicitly asks you to share the project configuration.

Then verify:

```
claude plugin list
claude plugin details dashboard-toolkit@team-marketplace
```

If the install summary requests activation, run:

```
/reload-plugins
```

Test `/dashboard-toolkit:review server.js` and verify both hooks on safe test files.

Exercise 5: Prove Reversibility (5 min)

```
/plugin disable dashboard-toolkit@team-marketplace
/plugin enable dashboard-toolkit@team-marketplace
/plugin uninstall dashboard-toolkit@team-marketplace
```

Reinstall at local scope after proving the lifecycle. Record which settings file changed for the selected scope.

FINISH MARKER - Success Checklist

- [] Inspected exact publisher, components, authority, data flow, and lifecycle before installation
- [] Recorded an evidence-based trust decision
- [] `claude plugin init dashboard-toolkit --with skills hooks` created a skills-directory plugin
- [] `claude plugin validate ... --strict` passed
- [] Marketplace uses `./plugins/dashboard-toolkit` inside its own repository
- [] Plugin installed at intentional scope and activated successfully
- [] Namespaced review skill and hooks work
- [] Disable, enable, uninstall, and reinstall paths were proven

If Stuck

Problem	Recovery
Scaffold not in current folder	Expected: it is under <code>~/ .claude/skills/dashboard-toolkit/</code>
Strict validation fails	Fix every schema error or warning before publication
Marketplace source not found	Keep plugin under <code>team-marketplace/plugins/</code> and use a marketplace-relative source
Plugin installed but inactive	Follow the install summary; run <code>/reload-plugins</code> when requested
Old component remains	Remove the installed copy, update the marketplace, and reinstall at the intended scope
Hook path breaks after install	Use <code>\${CLAUDE_PLUGIN_ROOT}</code> in plugin hook paths

Debrief Questions

1. Why is a marketplace a distribution mechanism rather than a security boundary?
2. Which component in your plugin carries the most authority?
3. What evidence would your organization require before moving this plugin into an approved internal marketplace?